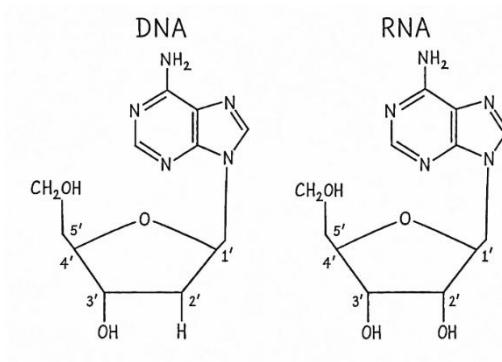
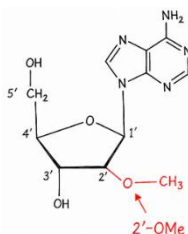


Furanose Sugar Modifications



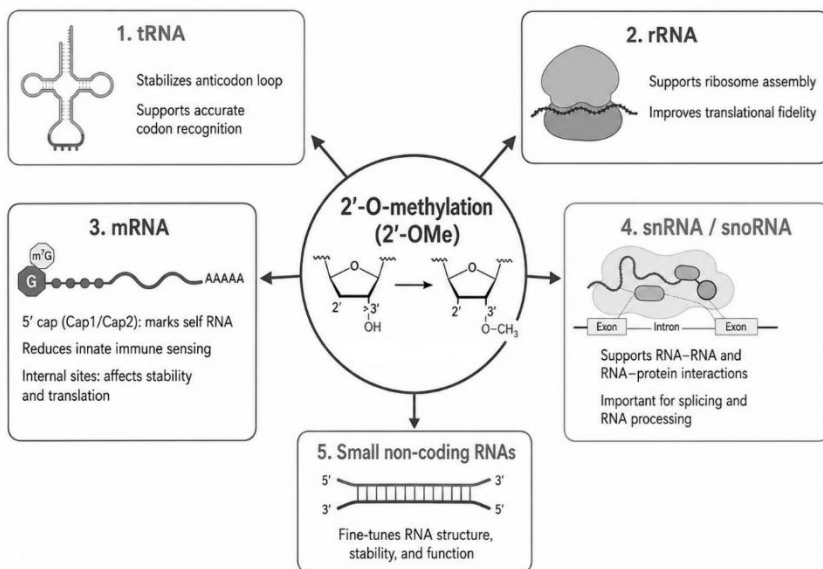
Sugar modifications are among the most important tools for converting fragile nucleic acids into durable, selective, and therapeutically useful molecules. By altering ribose or sugar-like scaffold, chemists can increase nuclease resistance, tune hybridization strength, reduce innate immune activation, improve strand selection, and shape how an oligonucleotide interacts with cellular machinery. Modifications such as 2'-O-methyl, 2'-fluoro, 2'-MOE, LNA, and GNA illustrate how small changes at or around the sugar can produce major effects on potency, safety, pharmacokinetics, and clinical performance. These chemistries form a central design layer in modern RNA therapeutics, especially for siRNAs, antisense oligonucleotides, splice-switching drugs.

2'-O-Methyl (2'-OMe)



Chemically, 2'-O-methylation replaces the ribose 2'-hydroxyl group with a methoxy substituent, increasing RNA conformational stability, nuclease resistance, and compatibility with endogenous RNA-recognition pathways while reducing innate immune stimulation. Endogenously, 2'-O-methylation is one of the most abundant and conserved RNA modifications,

occurring across multiple RNA classes including tRNA, rRNA, mRNA, snRNA, snoRNA, and small non-coding RNAs, where it fine-tunes RNA folding, stability, intermolecular recognition, and translational control (Dimitrova et al., 2019; Motorin and Helm, 2019). In tRNAs, 2'-OMe modifications help stabilize anticodon-loop structure and support accurate codon recognition, particularly at positions involved in decoding and wobble interactions. In rRNAs, 2'-OMe marks are installed largely through snoRNA-guided pathways and cluster around functionally important ribosomal regions, where they contribute to ribosome assembly, translational fidelity, and ribosome activity. In spliceosome snRNAs and snoRNAs, 2'-OMe supports RNA–RNA and RNA–protein interactions required for proper pre-mRNA splicing and ribonucleoprotein function.



In mRNAs, 2'-OMe occurs both at the 5' cap and internally: 2'-OMe generates cap1 and cap2 structures that mark host transcripts as “self” and suppress inappropriate innate immune sensing, while internal 2'-OMe can modulate translation elongation by influencing codon–anticodon decoding dynamics (Choi et al., 2018; Picard-Jean et al., 2018; Sendinc et al., 2023). Therapeutic RNA design adapts this same chemical logic for different purposes: in siRNAs, 2' ribose modifications can increase nuclease resistance, reduce immune activation, preserve RISC compatibility, and tune strand behavior.

Alnylam developed the first siRNA architecture for therapeutics; these made extensive use of the 2'OMe modifications where designs combined other modifications helped balance stability, potency, RISC loading, tissue exposure, and safety. First-generation Standard Template Chemistry (STC) used a fully modified 2'-OMe/2'-F duplex with limited

terminal phosphorothioate stabilization. Enhanced Stabilization Chemistry then increased terminal phosphorothioate linkages, raised the relative 2'-OMe content, and reduced total 2'-F content to improve metabolic stability and durability. Later advanced ESC and ESC+ designs further refined this architecture, including seed-region destabilizing modifications such as GNA to reduce off-target effects. Thus, Song et al. illustrate the mechanistic principle that modification position can influence strand selection and specificity, while Alnylam's STC-to-ESC progression shows how full-duplex chemical architectures were optimized for clinical GalNAc-siRNA performance.

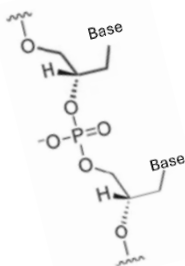
At Alnylam Pharmaceuticals, Janas and colleagues worked with Martin A. Maier and Jing-Tao Wu to evaluate the safety of 2'-F modifications in GalNAc-siRNA conjugates and to examine the transition from first-generation STC to ESC. Because some nucleoside analog drugs can cause mitochondrial toxicity, they tested whether 2'-F monomers released during siRNA metabolism posed similar risks. They compared revusiran, an STC GalNAc-siRNA, with ALN-TTRSC02/vutrisiran, an ESC GalNAc-siRNA with the same sequence, and measured 2'-F monomer release, tissue exposure, metabolic stability, and safety. Revusiran contained 22 2'-F modifications, reflecting the roughly balanced 2'OMe:2'-F composition of STC, approximately 22:22 or 1:1. In contrast, ALN-TTRSC02 contained only nine 2'-F modifications and a higher 2'-OMe content, giving an approximate 2'-OMe:2'-F ratio of 35:9, or about 4:1, assuming the same fully modified duplex length. This shift toward more 2'-OMe and less 2'-F, together with four additional terminal phosphorothioate linkages, reduced 2'-F metabolite exposure, improved metabolic stability, increased potency and duration of activity, and supported lower and less frequent dosing. Thus, ESC minimizes 2'-F metabolite exposure while preserving durable GalNAc-siRNA activity (Janas et al., 2019).

This broader principle now underlies clinically validated GalNAc-siRNA design: therapeutic siRNAs use combinations of 2'-OMe, 2'-F, phosphorothioate linkages, and ligand conjugation to protect the duplex, maintain RISC activity, reduce off-target and immune liabilities, and enable durable gene silencing. In approved and late-stage GalNAc-siRNA therapies such as givosiran, lumasiran, and inclisiran, these modification architectures support subcutaneous dosing, hepatocyte delivery, improved pharmacokinetics, and long-lasting target knockdown (Nair et al., 2014; Fitzgerald et al., 2017; Adams et al., 2020; Garrelfs et al., 2021; Raal et al., 2020).

This principle is central to Alnylam's enhanced stabilization chemistry, where 2'-OMe combined with 2'-fluoro modifications improves pharmacokinetics, reduces off-target effects, and enables durable gene

silencing in approved therapies such as givosiran, lumasiran, and inclisiran (Nair et al., 2014; Fitzgerald et al., 2017; Adams et al., 2020; Garrelfs et al., 2021; Raal et al., 2020).

Glycol Nucleic Acid (GNA)



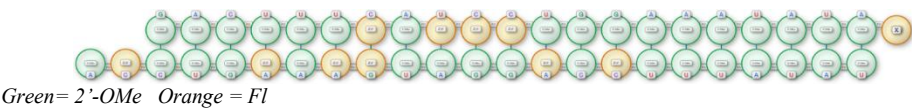
Unlike the preceding modifications, which are derived from naturally occurring RNA nucleosides, glycol nucleic acid (GNA) represents a synthetic nucleic acid analogue. GNA contains an acyclic three-carbon glycol backbone rather than the ribose sugar found in RNA and DNA, yet it retains the ability to form Watson–Crick base pairs and stable duplex structures (Zhang et al., 2005).

In RNA therapeutics, GNA has emerged as a valuable modification for optimizing siRNA performance. Strategic incorporation of (S)-GNA residues, particularly at position 7 within the antisense strand seed region, introduces controlled thermodynamic destabilization that reduces unintended seed-mediated off-target interactions while maintaining potent on-target gene silencing activity (Schlegel et al., 2022). GNA modifications also enhance resistance to nuclease degradation and can improve strand selection and loading into the RNA-induced silencing complex (RISC), contributing to overall therapeutic efficacy (Schlegel et al., 2017).

By enabling fine control over duplex stability and RNA structure, GNA improves specificity, reduces toxicity, and supports favorable pharmacological performance, making it an increasingly important component of next-generation siRNA therapeutics.

The success of RNA therapeutics has been driven in large part by advances in RNA chemistry. Naturally occurring modifications such as m⁵C, pseudouridine, and inosine provide mechanisms for enhancing RNA stability, regulating translation, reducing immunogenicity, and enabling programmable RNA editing, while synthetic analogues such as GNA offer

Lumasiran (Oxlumo)



Lumasiran (Oxlumo) is administered via subcutaneous and is an *ESC GalNAc-conjugated* siRNA, it is designed for targeted delivery to hepatocytes following systemic administration, enabling RNA interference of the HAO1 gene in the liver. Efficacy was successfully achieved in the phase 3 ILLUMINATE-A trial, where Lumasiran caused a 65% mean reduction in 24-hour urinary oxalate compared to a 12% reduction in the placebo group; this is a statistically significant difference; 54% ($p<0.0001$) (FDA, 2020). Approximately half of treated patients reached normalization of urinary oxalate levels by Month 6, indicating a clinically meaningful biochemical response (FDA, 2020). Consistent results were observed in younger patients in the ILLUMINATE-B study, where a 71% reduction in urinary oxalate (measured as oxalate-to-creatinine ratio) was reported, supporting efficacy across age groups (FDA, 2020). Across studies, reductions in urinary oxalate were rapid, occurring within the first few months of treatment and sustained over the study duration (FDA, 2020). The data demonstrates that Lumasiran provides substantial and consistent reductions in urinary oxalate (surrogate endpoint), supporting its efficacy in the treatment of primary hyperoxaluria type (FDA, 2020).

	siRNA	Target / indication
STC	Revusiran / ALN-TTRsc	TTR; ATTR amyloidosis
STC	ALN-AT3 / fitusiran	Antithrombin; hemophilia
ESC	ALN-TTRSC02 / vutrisiran	TTR; ATTR amyloidosis
ESC	Givosiran / GIVLAARI / ALN-AS1	ALAS1; acute hepatic porphyria
ESC	Cemdisiran / ALN-CC5	Complement C5; complement-mediated diseases
ESC	Lumasiran / ALN-GO1	HAO1; primary hyperoxaluria type 1
ESC	Inclisiran / ALN-PCSSc	PCSK9; hypercholesterolemia
ESC+	ALN-AAT02	AAT; alpha-1 liver disease
ESC+	ALN-HBV02 / VIR-2218	HBV
ESC+	ALN-AGT	Angiotensinogen; hypertension

Revusiran highlights an important disconnect between preclinical toxicology, early clinical tolerability, and late-stage clinical outcome. Revusiran was a first-generation GalNAc–siRNA in which the GalNAc ligand was attached to the 3' end of the sense strand; the duplex contained 22 2'-OMe and 22 2'-F residues, with two phosphorothioate linkages at the 3' end of the antisense strand to increase nuclease resistance (Sutherland et al., 2020). This high 2'-F content raised a reasonable safety question because some fluorinated nucleoside analogues have caused mitochondrial toxicity in humans. The classic example is FIAU, a fluorinated antiviral nucleoside analogue developed for hepatitis B, which caused severe multisystem toxicity, liver failure, lactic acidosis, and mitochondrial injury after repeat dosing. Mechanistically, FIAU toxicity was linked to mitochondrial dysfunction, including abnormal swollen hepatic mitochondria with disrupted cristae. Because revusiran metabolism could theoretically release 2'-F nucleoside or nucleotide monomers, Sutherland et al. specifically examined whether chronic exposure produced a similar mitochondrial injury pattern (Sutherland et al., 2020).

The nonclinical data did not show a FIAU-like toxicity profile. In monkeys, revusiran produced no cardiovascular or respiratory effects at doses up to 100 mg/kg, no neurological abnormalities in repeat-dose studies, and no dose-limiting toxicity. In chronic monkey studies, the NOAEL was 200 mg/kg, the highest dose tested in the 39-week study, despite strong pharmacologic reductions in TTR, vitamin A, and thyroxine (Sutherland et al., 2020). Rats were more sensitive: the NOAEL was 30 mg/kg in both the 6- and 26-week repeat-dose studies, based mainly on reversible liver findings and associated clinical chemistry changes at higher doses, including hepatocellular vacuolation and modest elevations in ALT, AST, ALP, and GGT. These findings suggested species-specific sensitivity and liver adaptation or accumulation, but they were largely reversible and did not progress to overt organ failure (Sutherland et al., 2020). Revusiran also showed no genotoxicity and no carcinogenicity in a two-year rat study at doses up to 100 mg/kg. Importantly, liver ultrastructural assessments in the 26-week rat and 39-week monkey studies did not reveal mitochondrial damage. In the two-year rat carcinogenicity study, investigators did observe enlarged, elongated, ring-shaped, or cup-shaped mitochondria in liver and skeletal muscle, but these changes lacked the swollen morphology and cristae disruption typical of FIAU-associated mitochondrial toxicity and were not accompanied by functional evidence such as elevated plasma lactate (Sutherland et al., 2020).

Janas et al. further addressed the 2'-F concern by testing whether 2'-F metabolites from GalNAc-siRNAs behave like toxic nucleoside analogues. They found that 2'-F monomer exposure from revusiran and ALN-TTRSC02 was low and transient in rats and humans, and that 2'-F triphosphates were neither preferred substrates nor strong inhibitors of human polymerases. They also found no obligate or non-obligate chain termination, which argues against the kind of polymerase-mediated mitochondrial toxicity associated with some nucleoside analog drugs (Janas et al., 2019). Some in vitro assays did show modest reductions in cell viability and mitochondrial DNA in selected cell types at high 2'-F nucleoside concentrations, so the data do not prove that 2'-F monomers are biologically inert under all conditions. However, those effects generally occurred at concentrations above those achieved in vivo, and long-term rat studies did not show significant 2'-F monomer accumulation or clear mitochondrial functional injury (Janas et al., 2019; Sutherland et al., 2020).

Clinically, the early human data was also reassuring. In phase 1 healthy volunteers, Zimmermann et al., reported that single and multiple subcutaneous doses of revusiran were generally well tolerated, with injection-site reactions as the most common adverse events and no serious adverse events, treatment discontinuations, or clinically meaningful renal, hematologic, thyroid, cytokine, CRP, or urinalysis abnormalities. Multiple dosing produced approximately 90% mean serum TTR reduction, confirming potent GalNAc-mediated delivery and RNAi activity in humans (Zimmermann et al., 2017). The discrepancy emerged later in the phase 3 ENDEAVOUR trial in patients with cardiomyopathic hATTR amyloidosis, where development was stopped after an imbalance in deaths in the revusiran arm. Sutherland et al. note that no conclusive causal relationship was established, but a drug-related effect could not be excluded (Sutherland et al., 2020). Overall, the available evidence suggests that revusiran's phase 3 mortality signal was not clearly explained by intrinsic 2'-F mitochondrial toxicity. Instead, the mismatch between preclinical/phase 1 safety and ENDEAVOUR may reflect a combination of species differences, the use of healthy animals and healthy phase 1 volunteers, the older and medically fragile cardiomyopathy population in phase 3, advanced heart failure, disease-related vulnerability, and the relatively high-burden revusiran dosing regimen.